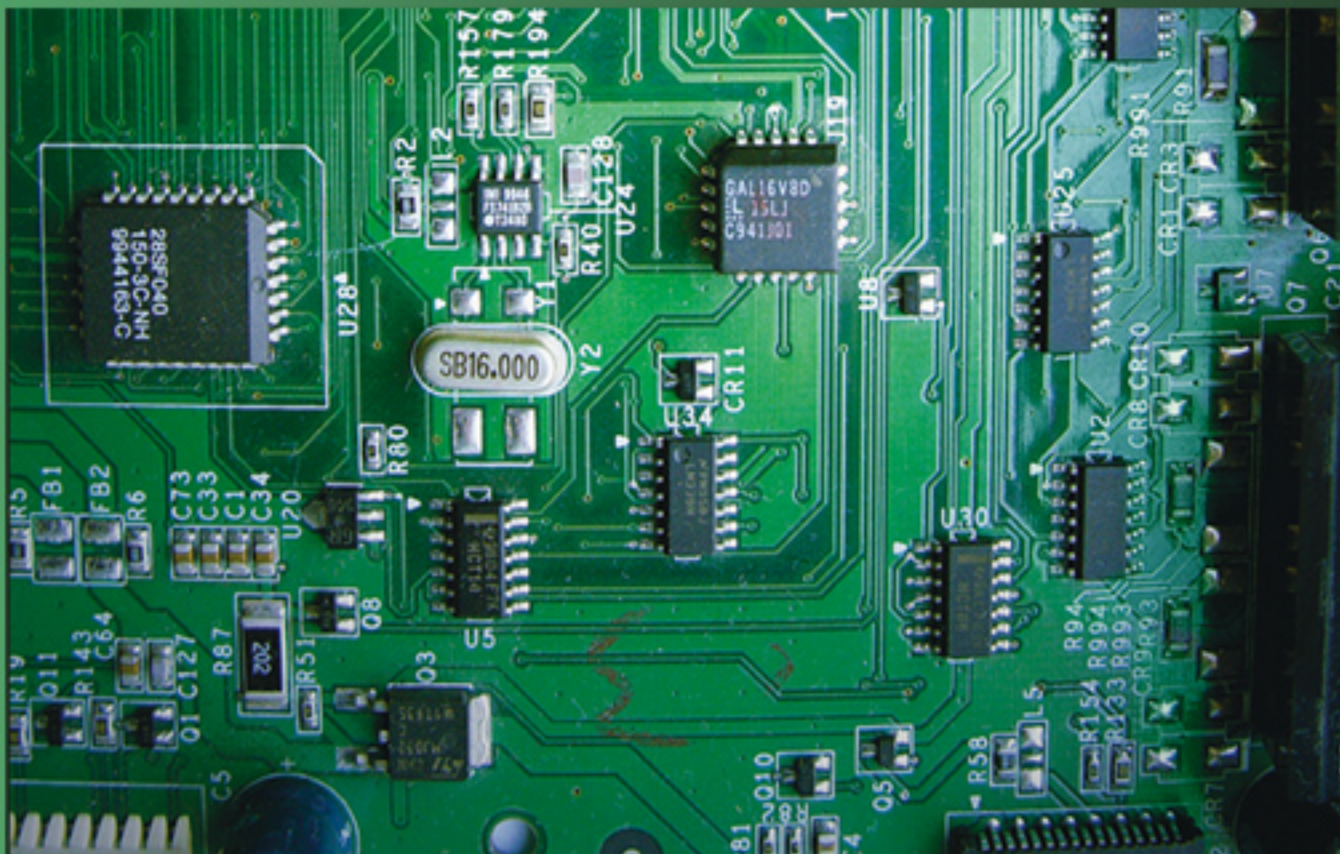
The Intel Microprocessors

8086/8088, 80186/80188, 80286, 80386, 80486 Pentium, Pentium Pro Processor, Pentium II, Pentium 4, and Core2 with 64-bit Extensions

Architecture, Programming, and Interfacing



EIGHTH EDITION

Barry B. Brey

PEARSON

Chapter 10: Memory Interface

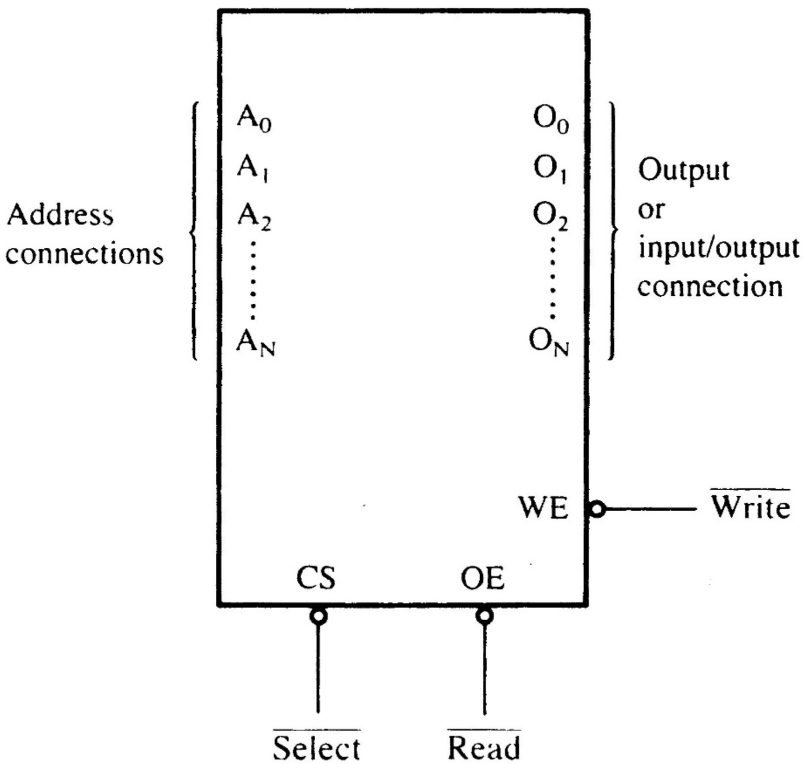
Introduction

- Simple or complex, every microprocessor-based system has a memory system.
- Almost all systems contain two main types of memory: **read-only memory** (ROM) and **random access memory** (RAM) or read/write memory.
- This chapter explains how to interface both memory types to the Intel family of microprocessors.

10–1 MEMORY DEVICES

- Before attempting to interface memory to the microprocessor, it is essential to understand the operation of memory components.
- In this section, we explain functions of the four common types of memory:
  - Read-only memory (ROM)
  - Flash memory (EEPROM)
  - Static random access memory (SRAM)
  - Dynamic random access memory (DRAM)

Memory Pin Connections



- address inputs
- data outputs or input/outputs
- some type of selection input
- at least one control input to select a read or write operation

Figure 10–1 A pseudo memory component illustrating the address, data, and control connections.

Address Connections

- Memory devices have address inputs to select a memory location within the device.
- Almost always labeled from A_0, the least significant address input, to A_n
  - where subscript n can be any value
  - always labeled as one less than total number of address pins
- A memory device with 10 address pins has its address pins labeled from A_0 to A_9.

Address Space (Main Memory: RAM)

- Address bus:10 bit □Address Space:1 Kbytes (2^{10})
- Address bus:11 bit □Address Space:2 Kbytes (2^{11})
- Address bus:16 bit □Address Space:64 KBytes (2^{16})
- Address bus:20 bit □Address Space:1 MBytes
  Address bus:32 bit □Address Space:4 GBytes
  Address bus:34 bit □Address Space:16GBytes
- Address bus:36 bit □Address Space:64GBytes
- Address bus:38 bit □Address Space:256GBytes
- Address bus:52 bit □Address Space:10^{52} Bytes

- The number of address pins on a memory device is determined by the number of memory locations found within it.
- Today, common memory devices have between 1K (1024) to 1G (1,073,741,824) memory locations.
  - with 4G and larger devices on the horizon
- A 1K memory device has 10 address pins.
  - therefore, 10 address inputs are required to select any of its 1024 memory locations

- It takes a 10-bit binary number to select any single location on a 1024-location device.
  - 1024 different combinations
  - if a device has 11 address connections, it has 2048 (2K) internal memory locations
- The number of memory locations can be extrapolated from the number of pins.

Data Connections

- All memory devices have a set of data outputs or input/outputs.
  - today, many devices have bidirectional common I/O pins
  - data connections are points at which data are entered for storage or extracted for reading
- Data pins on memory devices are labeled D_0 through D_7 for an 8-bit-wide memory device.

- An 8-bit-wide memory device is often called a **byte-wide** memory.
  - most devices are currently 8 bits wide,
  - some are 16 bits, 4 bits, or just 1 bit wide
- Catalog listings of memory devices often refer to memory locations times bits per location.
  - a memory device with 1K memory locations and 8 bits in each location is often listed as a 1K × 8 by the manufacturer
- Memory devices are often classified according to total bit capacity.

Selection Connections

- Each memory device has an input that selects or enables the memory device.
  - sometimes more than one
- This type of input is most often called a **chip select** ($\overline{\text{G2A}}$) **chip enable** ($\overline{\text{CE}}$) or simply **select** ($\overline{\text{S}}$) input.
- If more than one CE connection is present, all must be activated to read or write data.

Control Connections

- All memory devices have some form of control input or inputs.
  - ROM usually has one control input, while RAM often has one or two control inputs
- Control input often found on ROM is the **output enable** or **gate** connection, which allows data flow from output data pins.
- The $\overline{OE}$ connection enables and disables a set of three-state buffers located in the device and must be active to read data.

- RAM has either one or two control inputs.
  – if one control input, it is often called $R/\overline{W}$
- If the RAM has two control inputs, they are usually labeled $\overline{WE}$ (or $\overline{W}$), and $\overline{OE}$ (or $\overline{G}$).
  – **write enable** must be active to perform memory write, and $\overline{OE}$ active to perform a memory read
  – when the two controls are present, they must never both be active at the same time
- If both inputs are inactive, data are neither written nor read.
  – the connections are at their high-impedance state

ROM Memory

- **Read-only memory** (ROM) permanently stores programs/data resident to the system.
  - and must not change when power disconnected
- Often called **nonvolatile memory**, because its contents *do not* change even if power is disconnected.
- A device we call a ROM is purchased in mass quantities from a manufacturer.
  - programmed during fabrication at the factory

- The EPROM (**erasable programmable read-only memory**) is commonly used when software must be changed often.
- An EPROM is programmed in the field on a device called an EPROM programmer.
- Also erasable if exposed to high-intensity ultraviolet light.
  - depending on the type of EPROM

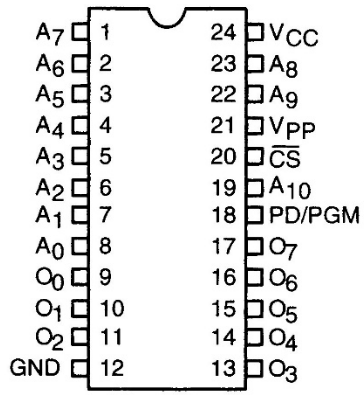
- PROM memory devices are also available, although they are not as common today.
- The PROM (**programmable read-only memory**) is also programmed in the field by burning open tiny NI-chrome or silicon oxide fuses.
- Once it is programmed, it cannot be erased.

- A newer type of **read-mostly memory** (RMM) is called the **flash memory**.
  - also often called an EEPROM (**electrically erasable programmable ROM**)
  - EAROM (**electrically alterable ROM**)
  - or a NOVRAM (**nonvolatile RAM**)
- Electrically erasable in the system, but they require more time to erase than normal RAM.
- The flash memory device is used to store setup information for systems such as the video card in the computer.

- Flash has all but replaced the EPROM in most computer systems for the BIOS.
  - some systems contain a password stored in the flash memory device
- Flash memory has its biggest impact in memory cards for digital cameras and memory in MP3 audio players.
- Figure 10–2 illustrates the 2716 EPROM, which is representative of most common EPROMs.

Figure 10–2 The pin-out of the 2716, 2K × 8 EPROM. (Courtesy of Intel Corporation.)

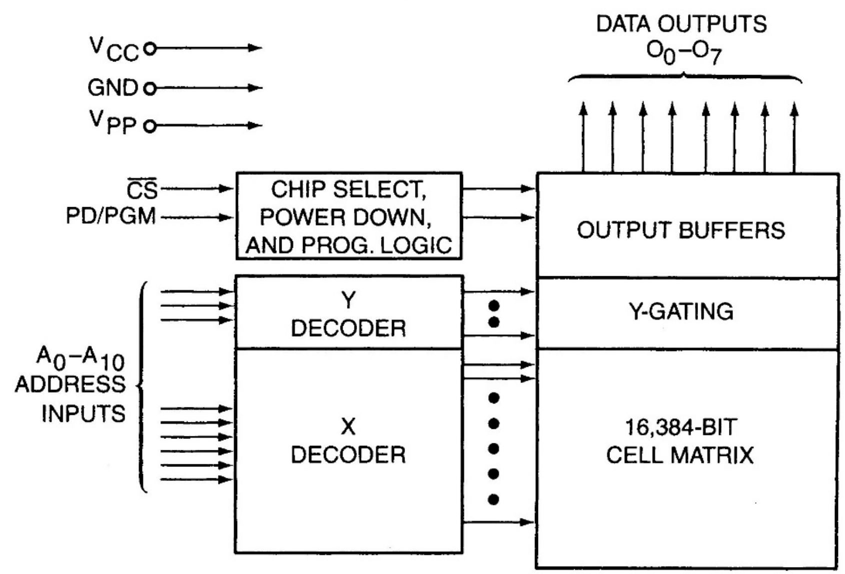
PIN CONFIGURATION



MODE SELECTION

PINS / MODE	PD/PGM (18)	$\overline{CS}$ (20)	V_{PP} (21)	V_{CC} (24)	OUTPUTS (9-11, 13-17)
Read	V_{IL}	V_{IL}	+5	+5	D_{OUT}
Deselect	Don't care	V_{IH}	+5	+5	High Z
Power Down	V_{IH}	Don't care	+5	+5	High Z
Program	Pulsed V_{IL} to V_{IH}	V_{IH}	+25	+5	D_{IN}
Program Verify	V_{IL}	V_{IL}	+25	+5	D_{OUT}
Program Inhibit	V_{IL}	V_{IH}	+25	+5	High Z

BLOCK DIAGRAM



PIN NAMES

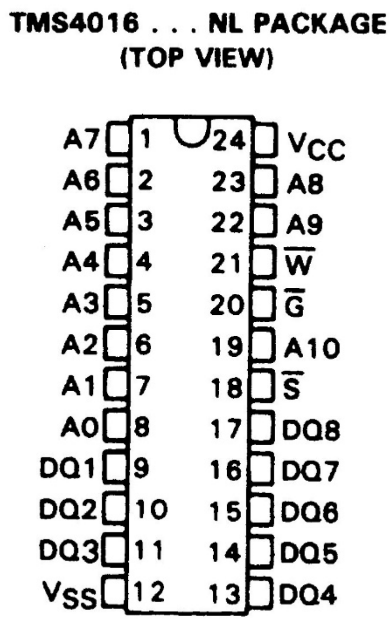
A_0–A_{10}	ADDRESSES
PD/PGM	POWER DOWN/PROGRAM
$\overline{CS}$	CHIP SELECT
O_0–O_7	OUTPUTS

Static RAM (SRAM) Devices

- Static RAM memory devices retain data for as long as DC power is applied.
- Because no special action is required to retain data, these devices are called **static memory**.
  - also called **volatile memory** because they will not retain data without power
- The main difference between ROM and RAM is that RAM is written under normal operation, whereas ROM is programmed outside the computer and normally is only read.

- Fig 10–4 illustrates the 4016 SRAM,
  - a 2K × 8 read/write memory
- This device is representative of all SRAM devices.
  - except for the number of address and data connections.
- The control inputs of this RAM are slightly different from those presented earlier.
  - however the control pins function exactly the same as those outlined previously
- Found under part numbers 2016 and 6116.

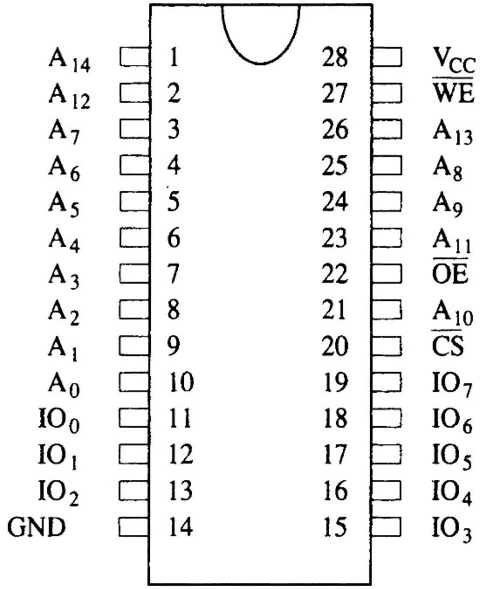
Figure 10–4 The pin-out of the TMS4016, 2K × 8 static RAM (SRAM). (Courtesy of Texas Instruments Incorporated.)



PIN NOMENCLATURE	
A0 – A10	Addresses
DQ1 – DQ8	Data In/Data Out
$\overline{G}$	Output Enable
$\overline{S}$	Chip Select
V_{CC}	+5-V Supply
V_{SS}	Ground
$\overline{W}$	Write Enable

- SRAM is used when the size of the read/write memory is relatively small
- today, a small memory is less than 1M byte

Figure 10–6 Pin diagram of the 62256, 32K × 8 static RAM.



PIN FUNCTION

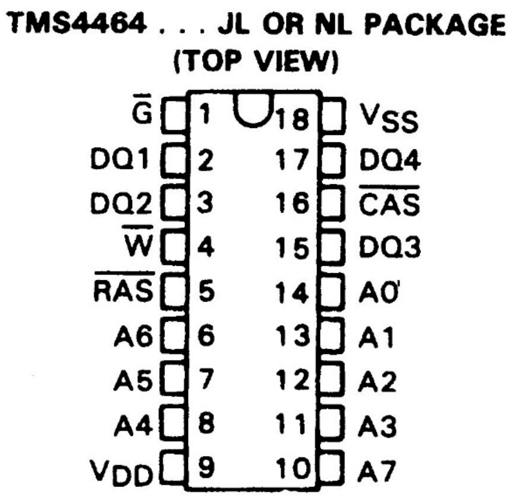
A_0 - A_{14}	Addresses
IO_0 - IO_7	Data connections
$\overline{CS}$	Chip select
$\overline{OE}$	Output enable
$\overline{WE}$	Write enable
V_{CC}	+5V Supply
GND	Ground

Dynamic RAM (DRAM) Memory

- Available up to 256M × 8 (2G bits).
- DRAM is essentially the same as SRAM, except that it retains data for only 2 or 4 ms on an integrated capacitor.
- After 2 or 4 ms, the contents of the DRAM must be completely rewritten (*refreshed*).
  - because the capacitors, which store a logic 1 or logic 0, lose their charges

- In DRAM, the entire contents are refreshed with 256 reads in a 2- or 4-ms interval.
  - also occurs during a write, a read, or during a special refresh cycle
- DRAM requires so many address pins that manufacturers multiplexed address inputs.
- Figure 10–7 illustrates a 64K × 4 DRAM, the TMS4464, which stores 256K bits of data.
  - note it contains only eight address inputs where it should contain 16—the number required to address 64K memory locations

Figure 10–7 The pin-out of the TMS4464, 64K × 4 dynamic RAM (DRAM). (Courtesy of Texas Instruments Incorporated.)



(a)

PIN NOMENCLATURE	
A0-A7	Address Inputs
$\overline{CAS}$	Column Address Strobe
DQ1-DQ4	Data-In/Data-Out
$\overline{G}$	Output Enable
$\overline{RAS}$	Row Address Strobe
V_{DD}	+ 5-V Supply
V_{SS}	Ground
$\overline{W}$	Write Enable

(b)

– 16 address bits can be forced into eight address pins in two 8-bit increments
– this requires two special pins: the **column address strobe** ($\overline{CAS}$) and **row address strobe** ($\overline{RAS}$)

10–2 ADDRESS DECODING

- In order to attach a memory device to the microprocessor, it is necessary to decode the address sent from the microprocessor.
- Decoding makes the memory function at a unique section or partition of the memory map.
- Without an address decoder, only one memory device can be connected to a microprocessor, which would make it virtually useless.

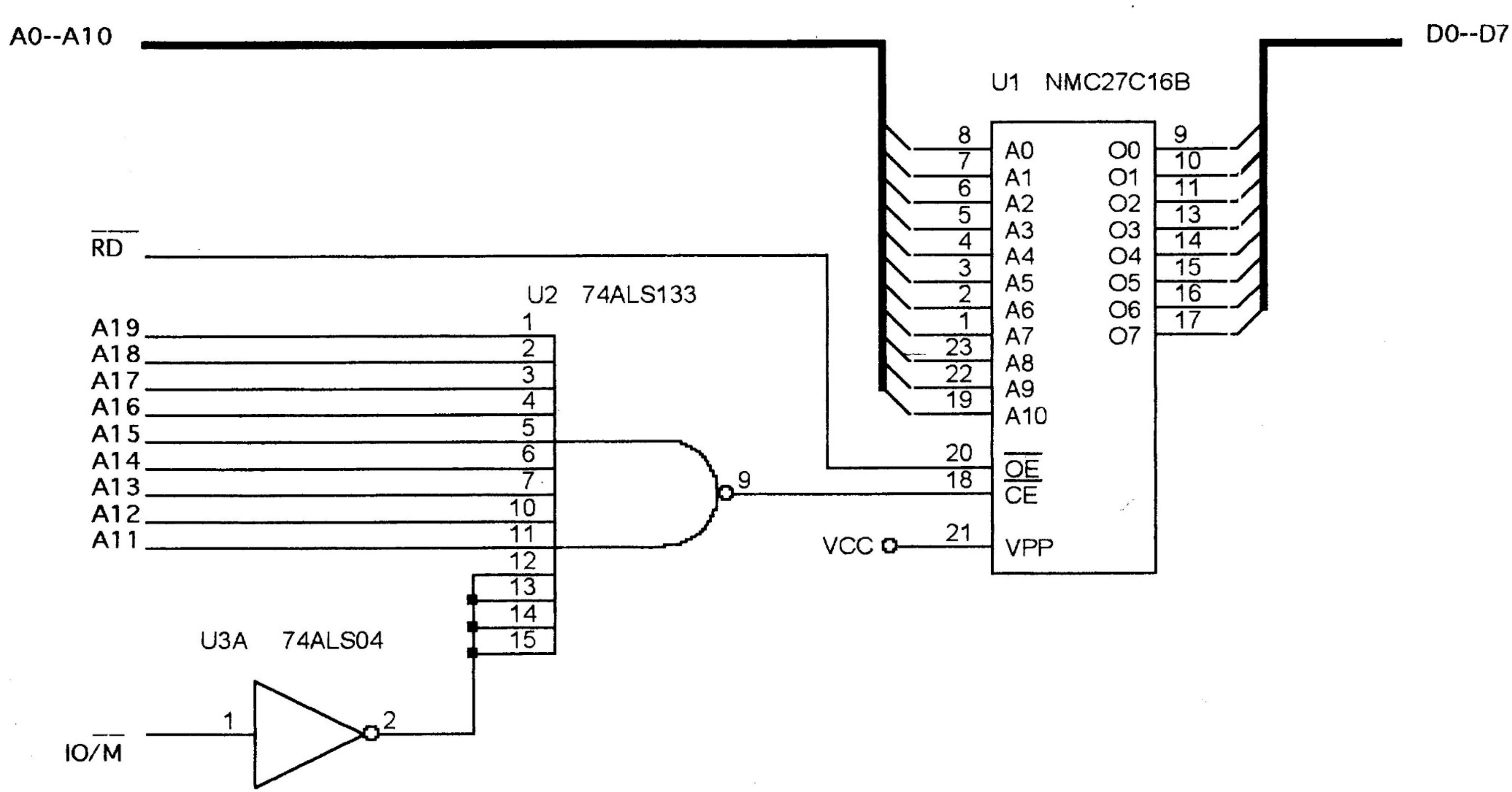
Why Decode Memory?

- The 8088 has 20 address connections and the 2716 EPROM has 11 connections.
- The 8088 sends out a 20-bit memory address whenever it reads or writes data.
  - because the 2716 has only 11 address pins, there is a mismatch that must be corrected
- The decoder corrects the mismatch by decoding address pins that do not connect to the memory component.

Simple NAND Gate Decoder

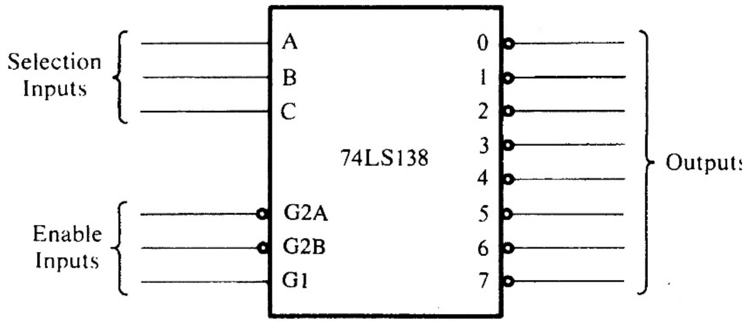
- When the 2K × 8 EPROM is used, address connections A_{10}–A_0 of 8088 are connected to address inputs A_{10}–A_0 of the EPROM.
  - the remaining nine address pins (A_{19}–A_{11}) are connected to a NAND gate decoder
- The decoder selects the EPROM from one of the 2K-byte sections of the 1M-byte memory system in the 8088 microprocessor.
- In this circuit a NAND gate decodes the memory address, as seen in Figure 10-13.

Figure 10–13 A simple NAND gate decoder that selects a 2716 EPROM for memory location FF800H–FFFFFH.



- If the 20-bit binary address, decoded by the NAND gate, is written so that the leftmost nine bits are 1s and the rightmost 11 bits are don't cares (X), the actual address range of the EPROM can be determined.
  - a *don't care* is a logic 1 or a logic 0, whichever is appropriate
- Because of the excessive cost of the NAND gate decoder and inverters often required, this option requires an alternate be found.

The 3-to-8 Line Decoder (74LS138)



Inputs						Outputs							
Enable			Select										
$\overline{G2A}$	$\overline{G2B}$	G1	C	B	A	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$
1	X	X	X	X	X	1	1	1	1	1	1	1	1
X	1	X	X	X	X	1	1	1	1	1	1	1	1
X	X	0	X	X	X	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	0

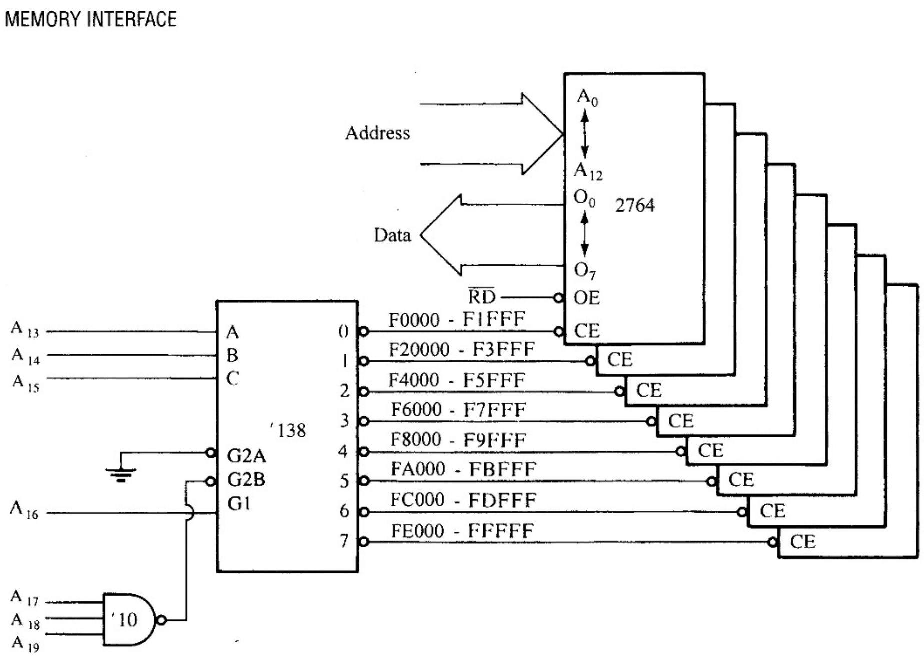
– a common integrated circuit decoder found in many systems is the 74LS138 3-to-8 line decoder.

Figure 10–14 The 74LS138 3-to-8 line decoder and function table.

Sample Decoder Circuit

- The outputs of the decoder in Figure 10–15, are connected to eight different 2764 EPROM memory devices.
- The decoder selects eight 8K-byte blocks of memory for a total capacity of 64K bytes.
- This figure also illustrates the address range of each memory device and the common connections to the memory devices.

Figure 10–15 A circuit that uses eight 2764 EPROMs for a 64K × 8 section of memory in an 8088 microprocessor-based system. The addresses selected in this circuit are F0000H–FFFFFH.



- all address connections from the 8088 are connected to this circuit.
- the decoder's outputs are connected to the CE inputs of the EPROMs,
- the RD signal from the 8088 is connected to the OE inputs of the EPROMs

- In this circuit, a three-input NAND gate is connected to address bits A_{19}–A_{17}.
- When all three address inputs are high, the output of this NAND gate goes low and enables input $\overline{G2B}$ of the 74LS138.
- Input G1 is connected directly to A_{16}.
- In order to enable this decoder, the first four address connections (A_{19}–A_{16}) must all be high.

- Address inputs C, B, and A connect to microprocessor address pins A_{15}–A_{13}.
- These three address inputs determine which output pin goes low and which EPROM is selected whenever 8088 outputs a memory address within this range to the memory system.

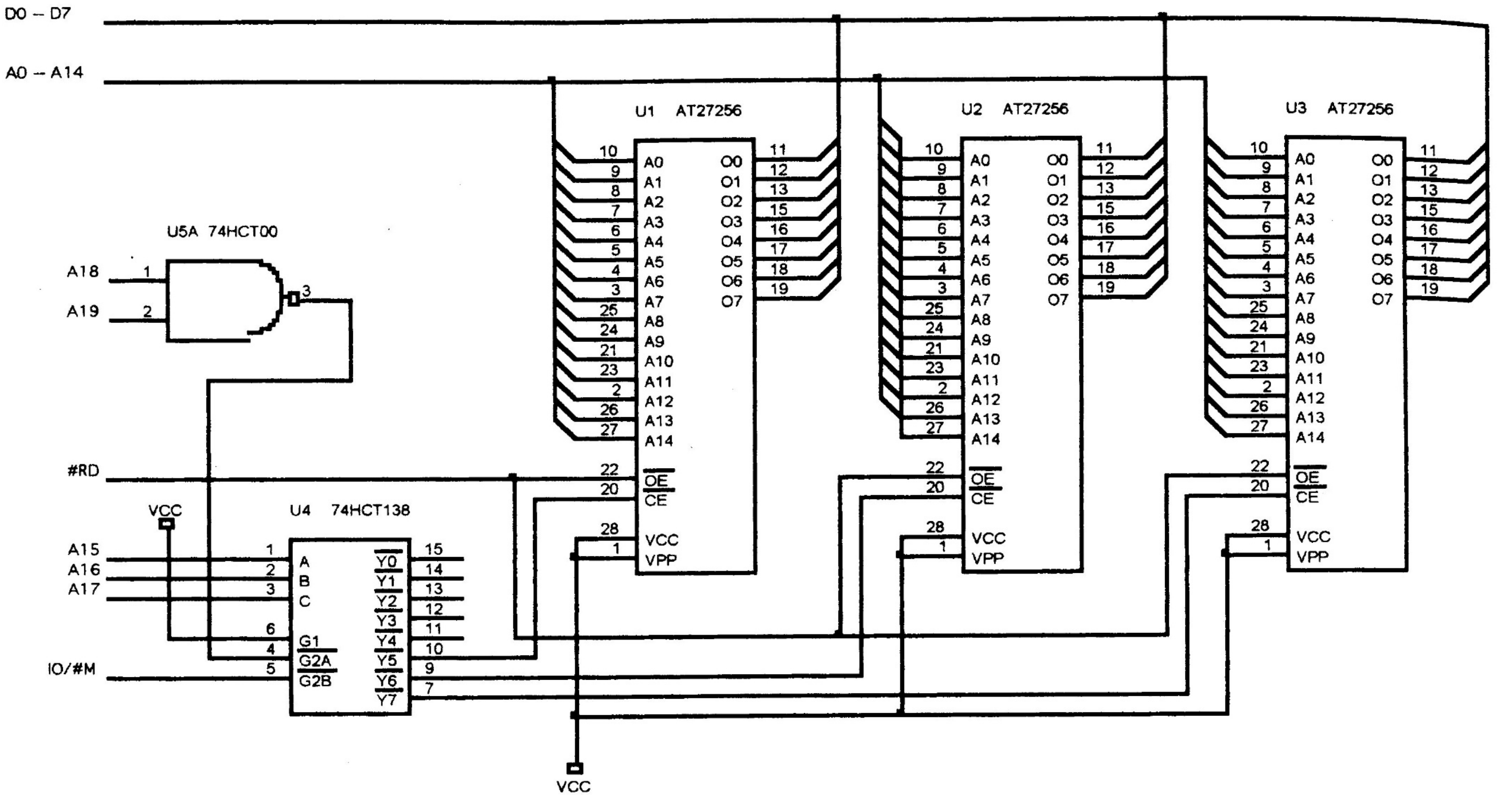
10–3 8088 and 80188 (8-bit) MEMORY INTERFACE

- 8088/80188 microprocessors have an 8-bit data bus, which makes them ideal to connect to common 8-bit memory devices available.
- For the 8088/80188 to function correctly with memory, however, the system must decode the address to select a memory component.

Interfacing EPROM to the 8088

- Fig 10–20 shows an 8088/80188 connected to three 27256 EPROMs, 32K × 8 devices.
  - 74HCT138 decoder in this illustration decodes three 32K × 8 blocks of memory for a total of 96K × 8 bits of physical address space for 8088/80188
- The decoder is selected for an address range that begins at E8000H and continues through location FFFFFH—the upper 96K bytes of memory.

Figure 10–20 Three 27256 EPROMs interfaced to the 8088 microprocessor.



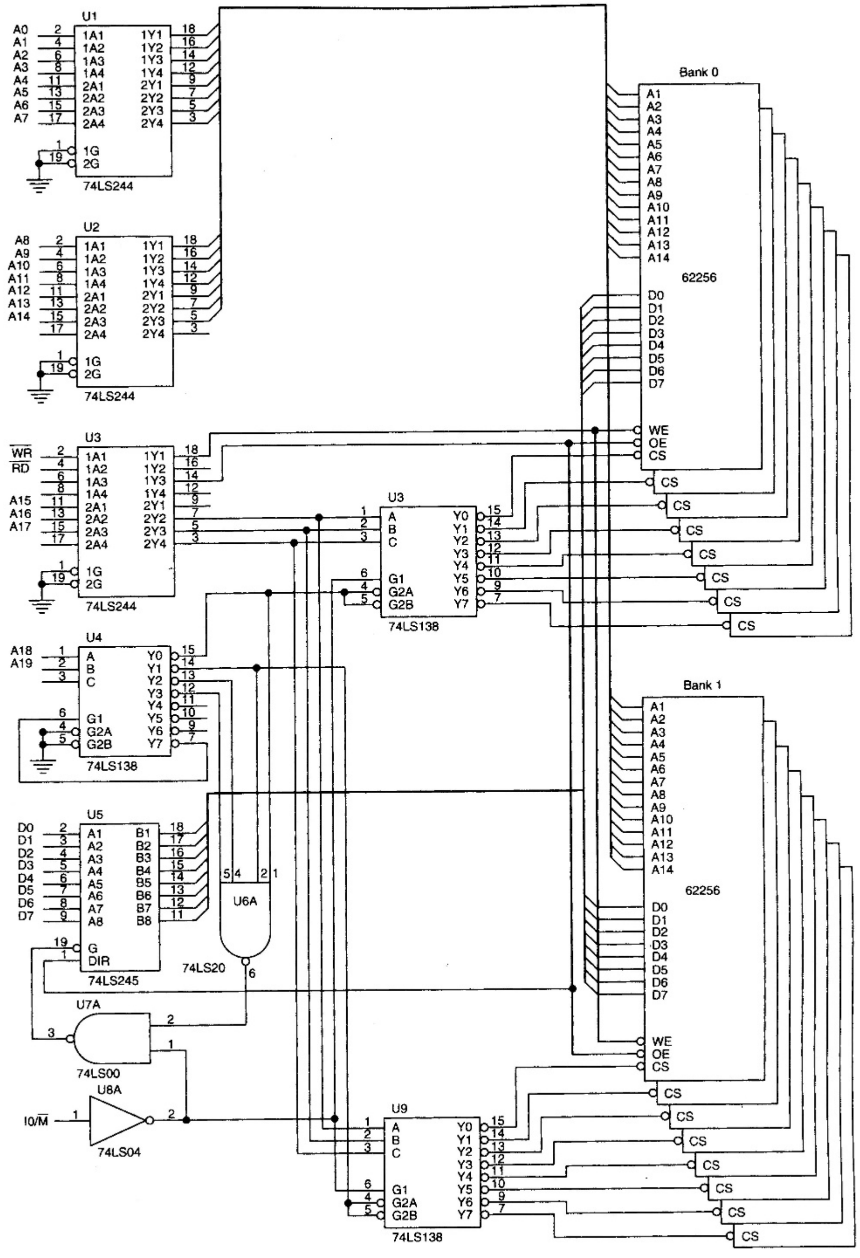
- In this circuit, U_1 is decoded at addresses E8000H–EFFFFH, U_2 is decoded at F0000H–F7FFFH, and U_3 at F8000H–FFFFFH.

Interfacing RAM to the 8088

- RAM is easier to interface than EPROM

- Fig 10–21 shows sixteen 62256 static RAMs interfaced to the 8088, beginning at 00000H.

Figure 10–21 A 512K-byte static memory system using 16 62255 SRAMs.



- this board uses two decoders to select 16 RAM components, and a third to select other decoders for the appropriate memory sections
- sixteen 32K RAMs fill memory from 00000H through 7FFFFH, for 512K bytes memory.